Trastuzumab emtansine versus taxane use for previously treated HER2-positive locally advanced or metastatic gastric or gastro-oesophageal junction adenocarcinoma (GATSBY): an international randomised, open-label, adaptive, phase 2/3 study

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See Comment page 562

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Summary

Background Although trastuzumab plus chemotherapy is the standard of care for first-line treatment of HER2-positive advanced gastric cancer, there is no established therapy in the second-line setting. In GATSBY, we examined the efficacy and tolerability of trastuzumab emtansine in patients previously treated for HER2-positive advanced gastric cancer (unresectable, locally advanced, or metastatic gastric cancer, including adenocarcinoma of the gastro-oesophageal junction).

Methods This is the final analysis from GATSBY, a randomised, open-label, adaptive, phase 2/3 study, done at 107 centres (28 countries worldwide). Eligible patients had HER2-positive advanced gastric cancer and progressed during or after first-line therapy. In stage one of the trial, patients were randomly assigned to treatment groups (2:2:1) to receive intravenous trastuzumab emtansine (3·6 mg/kg every 3 weeks or 2·4 mg/kg weekly) or physician's choice of a taxane (intravenous docetaxel 75 mg/m² every 3 weeks or intravenous paclitaxel 80 mg/m² weekly). In stage two, patients were randomly assigned to treatment groups (2:1) to receive the independent data monitoring committee (IDMC)-selected dose of trastuzumab emtansine (2·4 mg/kg weekly) or a taxane (same regimen as above). We used permuted block randomisation, stratified by world region, previous HER2-targeted therapy, and previous gastrectomy. The primary endpoint (overall survival) was assessed in the intention-to-treat population. This study is registered with ClinicalTrials.gov, number NCT01641939.

Findings Between Sept 3, 2012, and Oct 14, 2013, 70 patients were assigned to receive trastuzumab emtansine 3·6 mg/kg every 3 weeks, 75 to receive trastuzumab emtansine 2·4 mg/kg weekly, and 37 to receive a taxane in the stage 1 part of the trial. At the pre-planned interim analysis (Oct 14, 2013), the IDMC selected trastuzumab emtansine 2·4 mg/kg weekly as the dose to proceed to stage 2. By Feb 9, 2015, a further 153 patients had been randomly assigned to receive trastuzumab emtansine 2·4 mg/kg weekly and a further 80 to receive a taxane. At data cutoff, median follow-up was 17·5 months (IQR 12·1–23·0) for the trastuzumab emtansine 2·4 mg/kg weekly group and 15·4 months (9·2–18·1) in the taxane group. Median overall survival was 7·9 months (95% CI 6·7–9·5) with trastuzumab emtansine 2·4 mg/kg weekly and 8·6 months (7·1–11·2) with taxane treatment (hazard ratio 1·15, 95% CI 0·87–1·51, one-sided p=0·86). The trastuzumab emtansine 2·4 mg/kg group had lower incidences of grade 3 or more adverse events (134 [60%] of 224 patients treated with trastuzumab emtansine *vs* 78 [70%] of 111 patients treated with a taxane), and similar incidences of adverse events leading to death (eight [4%] *vs* four [4%]), serious adverse events (65 [29%] *vs* 31 [28%]), and adverse events leading to treatment discontinuation (31 [14%] *vs* 15 [14%]) than did taxane treatment. The most common grade 3 or more adverse events in the trastuzumab emtansine 2·4 mg/kg weekly group were anaemia (59 [26%]) and thrombocytopenia (25 [11%]) compared with neutropenia (43 [39%]), and anaemia (20 [18%]), in the taxane group. The most common serious adverse events were anaemia (eight [4%]), upper gastrointestinal haemorrhage (eight [4%]), pneumonia (seven [3%]), gastric haemorrhage (six [3%]), and gastrointestinal haemorrhage (five [2%]) in the trastuzumab emtansine 2·4 mg/kg weekly group compared with pneumonia (four [4%]), febrile neutropenia (four [4%]), anaemia (three [3%]), and neutropenia (three [3%]) in the taxane group.

Interpretation Trastuzumab emtansine was not superior to taxane in patients with previously treated, HER2-positive advanced gastric cancer. There is still an unmet need in this patient group and therapeutic options remain limited.

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Research in context

Evidence before this study

We did a systematic literature review, searching PubMed in addition to abstracts from the American Society of Clinical Oncology annual meeting, the Gastrointestinal Cancers Symposium, and the European Society for Medical Oncology biennial meeting. We used the search terms "gastric cancer", "HER2", and "trastuzumab emtansine" (full names and abbreviations), and relevant articles published up to Oct 31, 2015, were selected. Results were limited using these search terms, although the effectiveness of trastuzumab emtansine in pre-clinical models of HER2-positive gastric cancer was reported. We also examined the treatment landscape of second-line metastatic gastric cancer, using the resources mentioned above and the search terms "metastatic gastric cancer" and "HER2" (full names and abbreviations). Relevant articles published up to October, 2015, were selected. This search revealed publications reporting the activity of trastuzumab (phase 3 trial; first-line) and limited activity of lapatinib (phase 2 and 3 trials; second-line) in patients with metastatic gastric cancer. Thus, there is an unmet need for effective therapy for patients with HER2-positive advanced gastric cancer in a second-line setting.

Added value of this study

Here we report the result of a randomised phase 2/3 trial of trastuzumab emtansine in previously treated patients with advanced gastric cancer. The unique seamless adaptive design of GATSBY allowed instantaneous transition from the phase 2 dose-finding stage of the trial to the phase 3 stage, thereby accelerating the clinical development process and reducing the number of clinical trial participants required. Although GATSBY did not meet its primary endpoint of overall survival superiority in patients with previously treated HER2-positive advanced gastric cancer treated with trastuzumab emtansine compared with those treated with a taxane, this trial raises important questions about the biology of HER2-positive gastric cancers, particularly when compared with breast cancer. No clinical or biomarker subgroups were identified that showed a difference in overall survival in patients treated with trastuzumab emtansine compared with those treated with a taxane, and the secondary efficacy endpoint results were consistent with the those of the primary endpoint.

Implications of the available evidence

The findings from this trial suggest that trastuzumab emtansine is not superior to taxane treatment in patients with previously treated HER2-positive gastric cancer. Thus, there is still no HER2-targeted standard treatment regimen for this patient group and therapeutic options remain limited. This result suggests that HER2 overexpression functions differently in second-line advanced gastric cancer compared with metastatic breast cancer. HER2 overexpression in gastric cancer is often heterogeneous and it is not clear how focal HER2 expression might affect the activity of trastuzumab emtansine. Further research is needed to understand the role of HER2 overexpression in the biology of advanced gastric cancer.

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Introduction

Gastric cancer is the fifth most common cancer worldwide, with 952 000 new cases diagnosed in 2012.[1] It is the third most common cause of cancer death[1] and incidence varies geographically, with approximately half of all cases occurring in east Asia.[1]

The phase 3 Trastuzumab for Gastric Cancer (ToGA) study reported that 22·1% of patients with inoperable locally advanced, recurrent, or metastatic adenocarcinoma of the stomach or gastro-oesophageal junction (advanced gastric cancer) had tumour-specific overexpression of the *HER2* gene or amplified HER2 (immunohistochemistry score of 3+ or positive by fluorescence in-situ hybridisation [FISH]).[2] The incidence of HER2-positivity varied with histological subtype, the highest occurring in the intestinal subtypes and the lowest in diffuse or mixed subtypes.[2] HER2 overexpression is more common in adenocarcinomas located in the gastro-oesophageal junction than in those located in the gastric body.[2] In addition, HER2 overexpression in gastric cancer has a higher incidence of heterogeneity (focal staining) than breast cancer.[3] The level of *HER2* gene amplification has been shown to predict clinical outcome in patients with HER2-positive advanced gastric cancer who received trastuzumab.[4]

Trastuzumab is a HER2-targeted humanised monoclonal antibody that, in combination with chemotherapy, has been studied as a first-line treatment for patients with HER2-positive advanced gastric cancer. In the ToGA trial,[5] 584 patients received chemotherapy (capecitabine or fluorouracil plus cisplatin) plus trastuzumab or chemotherapy alone. Median overall survival was 13·8 months (95% CI 12–16) versus 11·1 months (10–13), respectively (hazard ratio [HR] 0·74; 95% CI 0·60–0·91, p=0·0046).[5] An exploratory post-hoc analysis showed that the overall survival benefit was substantially higher with trastuzumab plus chemotherapy in patients with high HER2 expression (defined as immunohistochemistry 2+ and a FISH-positive result, or immunohistochemistry 3+, regardless of FISH status) versus low HER2 expression (immunohistochemistry 0 or 1+ and FISH-positive).[5] Significant increases in progression-free survival, time to progression, and proportion of patients achieving an objective response were also reported in patients who received trastuzumab.[5] Based on these data, trastuzumab plus chemotherapy was approved for use in patients with previously untreated, HER2-positive, metastatic adenocarcinoma of the stomach or gastro-oesophageal junction who had not received previous anti-cancer treatment for their metastatic disease.

Although the benefit of adding HER2-targeted therapy (trastuzumab) to chemotherapy has been shown in the first-line setting for patients with advanced gastric cancer, this has not been established for second-line therapy, where clinical benefit has been shown with chemotherapy.[6] Results from three phase 3 studies showed that use of second-line docetaxel or irinotecan for advanced gastric cancer, compared with best supportive care, prolonged overall survival,[7–9] and a Japanese phase 3 trial showed similar overall survival benefit with either irinotecan or paclitaxel.[10] These results have established irinotecan, docetaxel, and paclitaxel each as chemotherapeutic options in the second-line setting. Recently, ramucirumab (an antivascular endothelial growth factor receptor 2 [VEGFR2] monoclonal antibody) also showed improved overall survival whether used alone or in combination with paclitaxel in the second-line setting for previously treated advanced gastric cancer.[11,12] The TyTAN study,[13] which enrolled Asian patients only, showed modest activity of second-line lapatinib plus paclitaxel for HER2-positive advanced gastric cancer, but did not demonstrate longer overall survival compared with patients treated with paclitaxel alone.[13] At present there are no clinical outcome data available from a global study with HER2-targeted therapy in the second-line advanced gastric cancer setting.[14–17]

See Online for appendix

Trastuzumab emtansine (T-DM1, Kadcyla; F Hoffmann-La Roche, Basel, Switzerland; Genentech, South San Francisco, CA, USA; ImmunoGen, Waltham, MA, USA), is an antibody–drug conjugate, comprised of trastuzumab linked by a stable linker to the tubulin inhibitor emtansine.[18] Intracellular release of cytotoxic emtansine-containing catabolites in HER2-positive breast cancer cells induces mitotic arrest and apoptosis.[18] Like trastuzumab, trastuzumab emtansine inhibits HER2-mediated signalling and shedding of the HER2 extracellular domain, and mediates antibody-dependent cellular cytotoxicity.[19,20] In patients with HER2-positive metastatic breast cancer who progressed after lines of HER2-targeted therapy, two phase 3 trials reported longer overall survival in patients treated with trastuzumab emtansine compared with patients treated with either lapatinib plus capecitabine[21] or physician's choice of treatment (which included chemotherapy, hormone therapy, or targeted therapy).[22,23] In EMILIA, a randomised phase 3 trial in patients with HER2-positive advanced breast cancer previously treated with trastuzumab and a taxane, patients treated with trastuzumab emtansine in the second-line setting had significantly longer median overall survival and progression-free survival compared with patients treated with lapatinib plus capecitabine. In addition, there was a lower incidence of grade 3 or more adverse events in the trastuzumab emtansine group than the lapatinib plus capecitabine group.[21] In TH3RESA, a phase 3 trial in patients who received at least two previous HER2-targeted therapies for HER2-positive metastatic breast cancer, patients treated with trastuzumab emtansine had longer median progression-free survival,[22] longer overall surival,[23] and fewer grade 3 or more adverse events[22,23] than patients treated with physician's choice of treatment. Trastuzumab emtansine (3·6 mg/kg every 3 weeks) is approved by the European Medicines Agency and the US Food and Drug Administration for the treatment of HER2-positive metastatic breast cancer in patients who had previously received trastuzumab and a taxane (separately or in combination).

Our preclinical data (appendix pp 1, 14–16), together with the findings from ToGA[5] and the phase 3 efficacy and safety data of trastuzumab emtansine in patients with metastatic breast cancer,[21–23] provided the rationale for assessing trastuzumab emtansine in patients with previously treated HER2-positive gastric cancer. Pharmacokinetic analyses from ToGA showed that, at the same starting dose, trastuzumab serum concentrations were lower in patients with metastatic gastric cancer than in historical data from patients with metastatic breast cancer.[24] Thus, GATSBY was designed as a phase 2/3 adaptive and seamless study: the phase 2 part aimed to examine two dosing regimens of trastuzumab emtansine and select the most appropriate dose for the phase 3 part, and the phase 3 part aimed to compare the efficacy and safety of the selected dose of trastuzumab emtansine with taxane therapy. The choice of single-agent taxane (either docetaxel or paclitaxel) as the control group was based on existing treatment guidelines for gastric cancer[25] and a global feasibility study. Here, we report the efficacy and safety findings of the selected trastuzumab emtansine dosing regimen compared with taxane dosing in patients with previously treated HER2-positive advanced gastric cancer.

Methods

Study design and participants

GATSBY was a three-arm, randomised, open-label, adaptive, phase 2/3 global study assessing the efficacy and safety of trastuzumab emtansine versus taxane treatment in patients with previously treated HER2-positive advanced gastric cancer (figure 1). The phase 2 part of the trial examined two dosing regimens of trastuzumab emtansine (2·4 mg/kg weekly compared with 3·6 mg/kg every 3 weeks) to allow selection of the most appropriate dose for assessment in the phase 3 part, which compared the efficacy and safety of the selected dose of trastuzumab emtansine with taxane treatment. Stage 1 of GATSBY began with the enrollment of the first patient and ended with the clinical cut-off for the regimen selection analysis; stage 2 included all patients recruited after the regimen selection decision and ended with the last patient out. The phase 3 part of the study included patients from both Stage 1 and Stage 2. Patients were enrolled from 107 centres in 28 countries worldwide.

Eligible patients were aged at least 18 years, had centrally tested HER2-positive, measurable and/or evaluable advanced gastric cancer (based on Response

Evaluation Criteria In Solid Tumors [RECIST] version 1.1), and had progressed either during or after first-line therapy with a combination of at least a platinum agent and a fluoropyrimidine given concurrently. Key eligibility criteria included an Eastern Cooperative Oncology Group performance status (ECOGPS) score of 0 or 1 and adequate organ function. Patients whose advanced gastric cancer progressed or recurred in 6 months or less after completion of adjuvant or neoadjuvant therapy could participate. In this instance, adjuvant or neoadjuvant therapy counted as one previous (first-line) therapy for advanced gastric cancer if it included a combination of at least a platinum agent and a fluoropyrimidine. Patients who experienced disease progression more than 6 months after the last dose of adjuvant or neoadjuvant therapy were required to receive first-line therapy for advanced gastric cancer and disease progression before enrolment. Patients must have experienced documented objective radiographic or pathological disease progression during or after first-line therapy for their disease. Patients were not eligible to participate if they had had another malignancy within the previous 5 years (with the exception of appropriately treated carcinoma in situ of the cervix, non-melanoma skin carcinoma, stage 1 uterine cancer, or other malignancies with an expected curative outcome), brain metastases (untreated, symptomatic, or required any radiation, surgery, or steroid therapy to control the symptoms of the brain metastasis within 1 month of randomisation), peripheral neuropathy (grade ≥2), cardiopulmonary dysfunction, or concurrent, serious, uncontrolled, or known infection with HIV or active hepatitis B virus or hepatitis C virus.

Patients who received previous treatment with HER2-targeted therapy were allowed to participate in the study. The last dose of chemotherapy or HER2-targeted therapy had to be at least 21 days before randomisation. Patients who received more than one prior line of therapy for advanced gastric cancer were excluded, along with those previously treated with trastuzumab emtansine or a taxane, and patients with clinically significant bleeding within the 30 days before enrolment. Patients who received previous treatment with trastuzumab, lapatinib, or pertuzumab were allowed to participate. Patients with a history of exposure to anthracyclines (maximum cumulative dose of >500 mg/m² or equivalent for doxorubicin or >720 mg/m² for epirubicin) were excluded. Treatment with any anti-cancer investigational drug within 21 days of the first administration of study treatment was not permitted.

HER2 status for study eligibility was determined with PATHWAY anti-HER2/neu (4B5) immunohistochemistry (Ventana Medical Systems, Tucson, AZ, USA) and the INFORM HER2 Dual in-situ hybridisation (ISH) assay (Ventana Medical Systems, Tucson, AZ, USA). Testing was performed at a central laboratory (Targos Molecular Pathology GmbH, Kassel, Germany) on either the primary or metastatic tumour, which was supplied as either a formalin-fixed paraffin-embedded tumour tissue block, as a partial block, or as freshly cut, unstained slides. Patients were eligible to participate in the study if on central laboratory testing their primary or metastatic tumours were positive for HER2 expression (immunohistochemistry 2+ and an ISH-positive result, or immunohistochemistry 3+, regardless of ISH status). Submission of new biopsy material for HER2 testing was not mandatory.

Essential laboratory tests to assess eligibility (within 28 days before randomisation) were an adequate haematological, renal, and hepatic function (neutrophil count ≥1500 cells/μL, platelet count ≥100 000 cells/μL, haemoglobin ≥9·0 g/dL, serum creatinine ≤1·5×the upper limit of normal [ULN], international normalised ratio and activated partial thromboplastin time <1·5×the ULN, serum aspartate aminotransferase [AST] and alanine aminotransferase [ALT] <2·5×the ULN, and serum total bilirubin <1·5×the ULN). Samples for haematology and serum chemistry were analysed at the study site's local laboratory and completed within 7 days of cycle 1, day 1. Laboratory assessments completed before each dose of study drug included haematology and serum chemistry assessments. Pregnancy testing for women of childbearing potential was done at baseline within 7 days of cycle 1, day 1, every three cycles, at study drug completion visit or early termination visit, and at approximately 3 and 6 months after the last dose of study drug. Urinalysis, international normalised ratio, and activated partial thromboplastin time were assessed at screening and repeated as clinically indicated.

Measurable and non-measurable disease was documented at screening and re-assessed at each subsequent tumour assessment. We did tumour assessments with

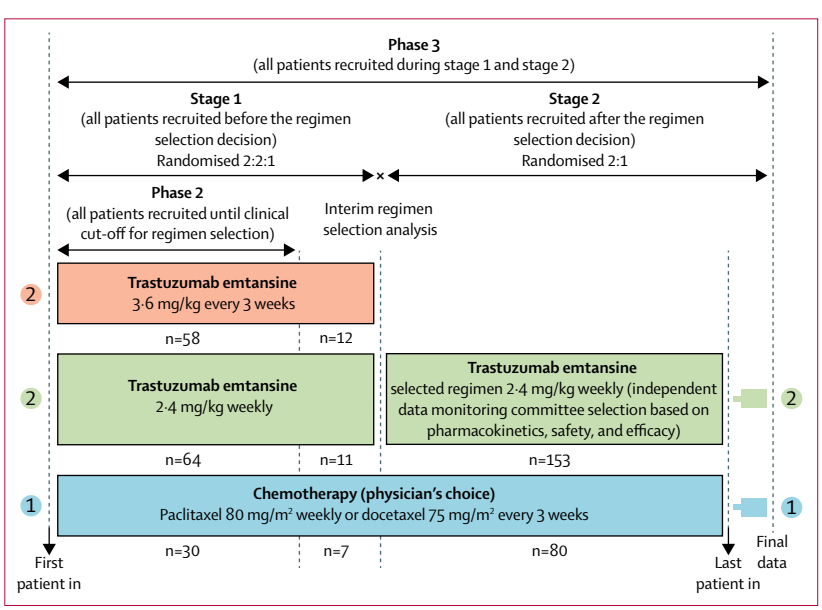

Figure 1: **GATSBY study design**

CT or MRI scans of the chest, abdomen, and pelvis every 6 weeks until disease progression, independently of the schedule of treatment administration or early treatment discontinuation. The same radiographic procedure used at screening was used throughout the study. Response assessments were determined by the investigator with RECIST (version 1.1).

GATSBY was conducted in accordance with the International Conference on Harmonisation (ICH) E6 Guidelines for Good Clinical Practice, the Declaration of Helsinki, and the ICH E2A guideline for expedited clinical safety data reporting. Each site was required to submit written documentation of protocol approval by the local ethics committee or institutional review board before initiating the study. All patients provided written informed consent.

Randomisation and masking

Patients were enrolled using a permuted block randomisation procedure with an interactive voice response system or interactive web response system (IxRS [IVRS or IWRS]; Almac Group, Craigavon, UK). The stratification factors were world region (Asia Pacific; western Europe, USA, and Canada; other [eastern Europe, Latin America, and remaining countries]), previous HER2-targeted therapy (yes *vs* no), and previous gastrectomy (yes *vs* no). Neither patients nor investigators were masked to treatment assignment; the sponsor remained masked to results from each treatment group during the study. During stage one of the trial, patients were randomly assigned to treatment groups (2:2:1) to receive either trastuzumab emtansine (3·6 mg/kg every 3 weeks or 2·4 mg/kg weekly) or the physician's choice of taxane (intravenous docetaxel [75 mg/m² every 3 weeks] or intravenous paclitaxel [80 mg/m² every week]). Stage one patients who were assigned to the regimen that was not selected for further evaluation continued to receive their assigned regimen and were not allowed to cross over. During stage two of the trial, patients were randomly assigned to treatment groups (2:1) to receive either the independent data monitoring committee (IDMC)-selected dose of trastuzumab emtansine (2·4 mg/kg weekly) or physician's choice of taxane treatment (either intravenous docetaxel [75 mg/m² every three weeks] or intravenous paclitaxel [80 mg/m² every week]).

Procedures

During stage one of the trial, patients received intravenous taxane treatment or trastuzumab emtansine 3·6 mg/kg every 3 weeks (the approved dose for patients with metastatic breast cancer) or 2·4 mg/kg weekly (high exposure, equivalent to 7·2 mg/kg in 3 weeks), with one regimen selected for the phase 3 part of the trial (figure 1).

Prior to regimen-selection analysis, serum samples for pharmacokinetic assessment of trastuzumab emtansine and total trastuzumab, and plasma samples for pharmacokinetic assessment of emtansine, were collected from approximately 40 patients in each of the trastuzumab emtansine-containing groups. The sample collection for pharmacokinetic analysis was not based on a statistical rationale, but on the number of patients initially planned to be enrolled in each trastuzumab emtansine group during the dose-selection phase. Trastuzumab emtansine and total trastuzumab concentrations in serum samples were quantified with validated enzyme-linked immunosorbent assays. Emtansine plasma concentrations were determined with a validated liquid chromatography tandem mass spectrometry (LC-MS/MS) method.

Patients treated with a taxane received either intravenous docetaxel 75 mg/m² every 3 weeks or paclitaxel 80 mg/m² weekly; one cycle was 21 days. Dosing for the every 3 weeks regimen occurred on day 1 of each cycle, whereas dosing for the weekly regimen occurred on days 1, 8, and 15. Taxane selection was made before randomisation at the physician's discretion. Patients were treated until progressive disease, intolerable toxicity, initiation of another anti-cancer therapy, patient and/or physician decision to discontinue study treatment, or study termination. Administration of trastuzumab emtansine, docetaxel, or paclitaxel could be delayed for up to 21 days (42 days from the last dose for the every 3 weeks regimen or 28 days from the last dose for the weekly regimen) to assess or treat adverse events. Once a dose reduction was made, there was no dose re-escalation allowed. If toxicities did not resolve within the authorised dose delay period, patients discontinued study treatment but continued to be followed for 30 days for adverse events and serious adverse events. General guidance for managing toxicities was provided in the study protocol (appendix p 3). After disease progression, patients were followed for survival.

Patients could discontinue the study drug or be withdrawn from the study at any time for reasons including, but not limited to, patient withdrawal of consent, any medical condition that the investigator or sponsor considered might jeopardise patient safety, if the investigator or sponsor determined it was in the best interest of the patient, and patient non-compliance.

Outcomes

The primary endpoint was overall survival. Secondary endpoints were progression-free survival, objective response rate, and duration of response (all by investigator's assessment), safety, patient-reported outcome measures, and pharmacokinetics. Clinical safety endpoints were the incidence, nature, and severity of adverse events, graded in accordance with National Cancer Institute Common Terminology Criteria for Adverse Events (NCI-CTCAE, version 4.03). Patient-reported outcomes included frequency of clinically significant treatment-related symptoms, time to advanced gastric cancer symptom progression, global health status,

quality of life, and health-related quality of life. The instruments used for reporting patient-reported outcomes were European Organisation for Research and Treatment of Cancer (EORTC) QLQ-C30, QLQ-STO22, and EQ-5D questionnaires. Exploratory endpoints included biomarker analyses of *HER2* gene amplification, HER2 and HER3 mRNA and protein expression, mutation analysis of the catalytic α-subunit of phosphatidylinositol 3-kinase (*PIK3CA*), and analysis of phosphatase and tensin homolog (PTEN) and cellular mesenchymal-epithelial transition (cMET) factor status.

Statistical analysis

Total sample size (412 patients) was calculated on the assumption that median overall survival in both trastuzumab emtansine groups would be 9 months versus 6 months in the taxane group; this corresponded to an HR of 0·67 between each trastuzumab emtansine group and the taxane group. The power of the study, defined as the probability that patients in the selected trastuzumab emtansine group would have significantly longer overall survival compared with the control group at the end of the study, was expected to be approximately 83%. As several of the groups defined by the stratification factors were expected to include fewer than five patients for the final analysis, the primary efficacy analysis was based on a non-stratified log-rank test. Median and 95% CIs were estimated with Kaplan-Meier survival methodology, and a non-stratified Cox proportional hazards model was used to estimate the HR (trastuzumab emtansine *vs* taxane), including 95% CIs. Overall survival was defined as the time from the date of randomisation to the date of death, regardless of cause of death. Patients for whom no death was reported before data cut-off were censored at the last known date they were alive. Patients with no post-baseline information were censored at one day after the date of randomisation. A one-sided inverse normal combination test was used to compare overall survival between the treatment groups, with a correction for the interim treatment selection due to the adaptive seamless design.[26] A sensitivity analysis, where patients were censored at the time of initiation of a new treatment, was performed for overall survival. SAS software (version 9.2; SAS, Cary, NC, USA) was used for statistical analyses.

The analyses for this type of trial design required adjustment for multiplicity because data from both study phases were included in the final analysis. The first step to adjust for multiplicity was calculation of two different p values: one p value was related to the group of patients

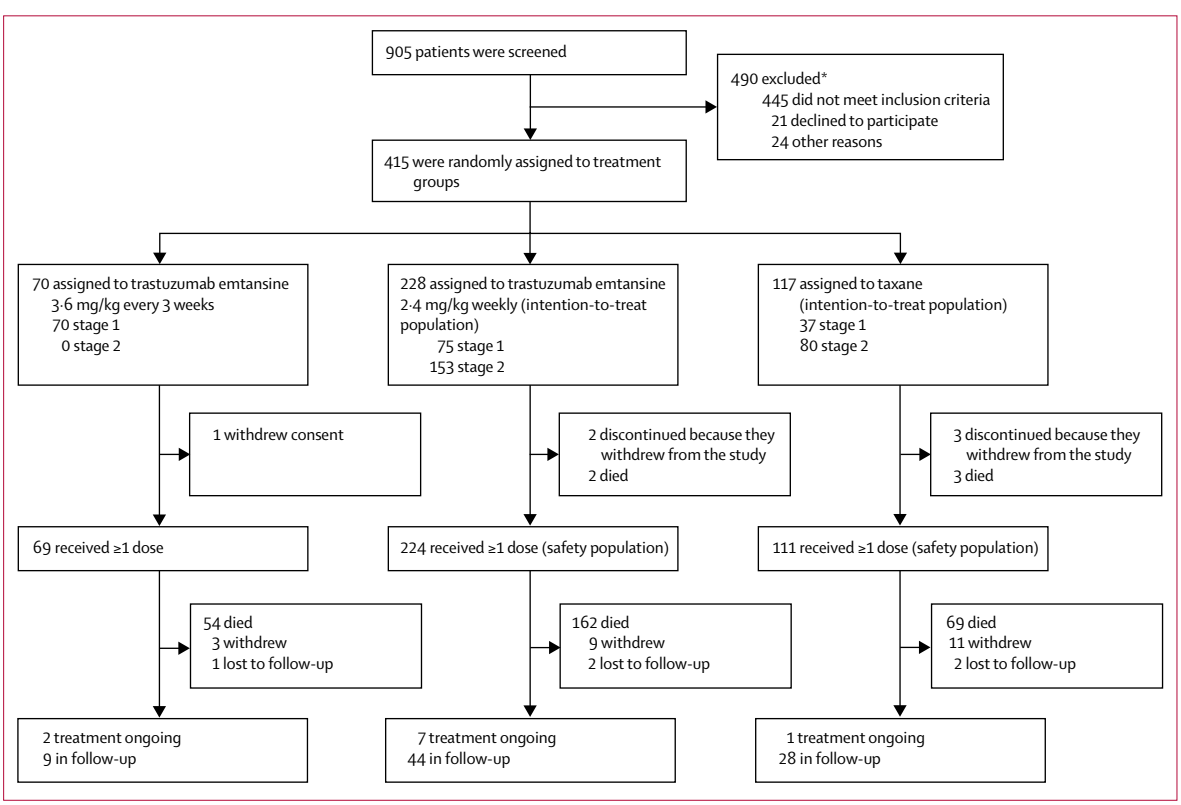

Figure 2: **Trial profile**

Data from the trastuzumab emtansine 3·6 mg/kg every 3 weeks group were analysed separately. *Other reasons for patient exclusion from GATSBY were unwillingness and inability to comply with scheduled visits, treatment plans, laboratory tests, and other study procedures (specifically including protocol-specified pharmacokinetic sampling), or assessment by the investigator that the patient would be unable or unwilling to comply with the requirements of the protocol.

	Taxane treatment (n=117)	Trastuzumab emtansine 2·4 mg/kg weekly (n=228)
Median age (years)	62·0 (27–80)	62·0 (19–79)
Age (years)		
<65	68 (58%)	140 (61%)
≥65	49 (42%)	88 (39%)
Sex		
Male	95 (81%)	177 (78%)
Female	22 (19%)	51 (22%)
Race		
White	57 (49%)	109 (48%)
Black	0	1 (<1%)
Asian	54 (46%)	105 (46%)
Other	6 (5%)	13 (6%)
World region		
Asia Pacific	53 (45%)	104 (46%)
Western Europe, USA, Canada	46 (39%)	88 (39%)
Other	18 (15%)	36 (16%)
ECOG PS		
0	43 (37%)	99 (43%)
1	73 (62%)	128 (56%)
2	1 (1%)	0
3	0	1 (<1%)
Primary site of disease		
Stomach	84 (72%)	151 (66%)
Gastro-oesophageal junction	33 (28%)	77 (34%)
Extent of disease at study entry		
Metastatic	113 (97%)	218 (96%)
Locally advanced	4 (3%)	10 (4%)
(Continued from previous column)		
Measurable disease	102 (87%)	204 (89%)
HER2 status*		
Immunohistochemistry 2+/ISH-positive	35 (30%)	78 (34%)
Immunohistochemistry 3+/ISH-positive	70 (60%)	125 (55%)
Immunohistochemistry 3+/ISH-negative	2 (2%)	6 (3%)
Immunohistochemistry 3+/ISH unknown	10 (9%)	18 (8%)
Previous anti-HER2 therapy	93 (79%)	174 (76%)
Trastuzumab	93 (79%)	173 (76%)
Pertuzumab	4 (3%)	18 (8%)
Lapatinib	0	1 (<1%)
Previous gastrectomy		
Yes	37 (32%)	66 (29%)
No	80 (68%)	162 (71%)
Histological subtype (Lauren classification)		
Intestinal	49 (42%)	88 (39%)
Diffuse	12 (10%)	43 (19%)
Mixed	5 (4%)	10 (4%)
Unknown	51 (44%)	87 (38%)

Data are n (%) or median (range). ECOG PS=Eastern Cooperative Oncology Group performance status. ISH=in-situ hybridisation. *The HER2 status for one patient in the trastuzumab emtansine group was confirmed centrally as HER2-negative following randomisation.

Table 1: **Baseline demographic and clinical characteristics in the intention-to-treat population**

randomly assigned to treatment groups before IDMC dose selection (including both trastuzumab emtansine regimens), and the other for the group of patients randomly assigned to treatment groups after this process. The two p values were then combined in an inverse normal combination test. A one-sided hypothesis testing framework was adopted to avoid conflicting directional effects from the two stages of the study, and to combine them into a single final directional test decision. This testing procedure allowed full control of the type I error for testing overall survival. The significance level of 2·5% was selected to account for using a one-sided p value.

The IDMC did data safety monitoring and regimen selection for trastuzumab emtansine (based on a pre-specified algorithm and after 100 patients had been treated and followed for a minimum of 12 weeks; the clinical cut-off date was Aug 10, 2013, and the interim analysis date was Oct 14, 2013) and reviewed pre-planned overall survival futility analysis (after 67 deaths occurred in the selected trastuzumab emtansine group and the control group across both stages of the trial). Efficacy was by intention-to-treat and safety analysis included all patients who had received at least one dose of the study treatment. The futility analysis took place in April, 2014; the IDMC recommended that the study continue without modification.

Progression-free survival, the proportion of patients achieving an objective response, and duration of response were based on the investigator's assessment of tumour response using RECIST (version 1.1). Progression-free survival was defined as the time from the date of randomisation to the date of first occurrence of progressive disease or death from any cause, whichever occurred first. In patients with measurable disease, objective response was defined as the achievement of a best overall response of partial response or complete response. Patients with a best overall response of stable disease, progressive disease, or those with missing data were considered non-responders.

Duration of response was defined as the time from the date when a clinical response (either complete response or partial response) was first documented to the date of progressive disease or death, whichever occurred first.

The phase 3 part of GATSBY included all data from the three treatment groups; the population used for confirmatory efficacy analysis was the intention-to-treat population of all patients randomly assigned to receive taxane or the selected dose of trastuzumab emtansine (2·4 mg/kg weekly; appendix p 1), and the safety population was all patients who received at least one dose of the study treatment. This trial is registered with ClinicalTrials.gov, number NCT01641939.

Role of the funding source

F Hoffmann-La Roche funded the study, provided the study drugs, and was involved in study design, protocol development, regulatory and ethics approvals, safety monitoring and reporting, data management, and data analysis and interpretation. Employees of the funder (MLH-Y, TvdH, BLA), and Silke Hoersch (KOEHLER eClinical, Germany) collected, managed, and analysed data, had access to the raw data, and were involved in the writing of the report. All authors reviewed and adapted the initial protocol, had full access to all study data, and interpreted and analysed the data. The corresponding author had full access to all of the data in the study and had final responsibility to submit for publication.

Results

Between Sept 3, 2012, and Oct 14, 2013, 70 patients were assigned to receive trastuzumab emtansine 3·6 mg/kg every 3 weeks, 75 to receive trastuzumab emtansine 2·4 mg/kg weekly, and 37 to receive a taxane in the stage 1 part of the trial (figure 2). At the pre-planned interim analysis (Oct 14, 2013; appendix pp 1, 4, 5, 17, 18), the IDMC selected trastuzumab emtansine 2·4 mg/kg weekly as the dose to proceed to stage 2. By June 30, 2015, a further 153 patients had been randomly assigned to receive trastuzumab emtansine 2·4 mg/kg weekly and a further 80 to receive a taxane (figure 2).

Baseline characteristics were generally well balanced (table 1, appendix p 3). Median duration of follow-up was similar between groups (17·5 months [IQR 12·1–23·0] for trastuzumab emtansine 2·4 mg/kg weekly and 15·4 months [IQR 9·2–18·1] for taxane treatment).

Trastuzumab emtansine did not show overall survival superiority (figure 3). Median overall survival was 7·9 months (95% CI 6·7–9·5) in the 2·4 mg/kg weekly trastuzumab emtansine group and 8·6 months (7·1–11·2) in the taxane group (HR 1·15, 95% CI 0·87–1·51, inverse normal combination test one-sided p=0·86). No clinical or biomarker subgroups showed overall treatment benefit with trastuzumab emtansine treatment (figure 4).

The results of the secondary efficacy endpoint analyses were consistent with those of the primary endpoint; although trastuzumab emtansine showed anti-tumour activity, it did not prolong progression-free survival compared with taxane treatment (figure 5). Median progression-free survival was 2·7 months (95% CI 1·6–2·7) in the trastuzumab emtansine 2·4 mg/kg weekly group and 2·9 months (2·8–4·0) in the taxane group (HR 1·13, 0·89–1·43, two-sided p=0·31). No clinical or biomarker subgroups showed prolonged progression-free survival with trastuzumab emtansine treatment. For patients with measurable disease at baseline, no difference in objective response was noted between treatment groups (table 2). In patients with a response, median duration of response was similar between treatment groups (table 2). Patient-reported outcomes were similar between treatment groups (data not shown). The median time to advanced gastric cancer symptom progression was 1·5 months (95% CI 1·4–1·6) in the trastuzumab emtansine 2·4 mg/kg weekly group and 1·6 months (1·4–2·2) in the taxane group (HR 0·98, 95% CI 0·79–1·26; data not shown).

Overall, at the time of clinical data cut-off, median treatment durations were 2·0 months for docetaxel (range 0–9 months; median dose received per infusion 74·8 mg/m²), 2·8 months for paclitaxel (range 0–11 months; median dose received per infusion 79·8 mg/m²; appendix p 5). Median time on trastuzumab emtansine 2·4 mg/kg weekly was 1·8 months (range 0–19 months; median dose received of 2·4 mg/kg; appendix p 5). Fewer patients receiving trastuzumab emtansine 2·4 mg/kg weekly (26 [12%] of 224 patients) required a dose reduction than patients receiving taxanes (17 [25%] of 69 patients who received docetaxel and ten [24%] of 41 patients who received paclitaxel; one patient received both paclitaxel and docetaxel and was therefore not included in the exposure summaries).

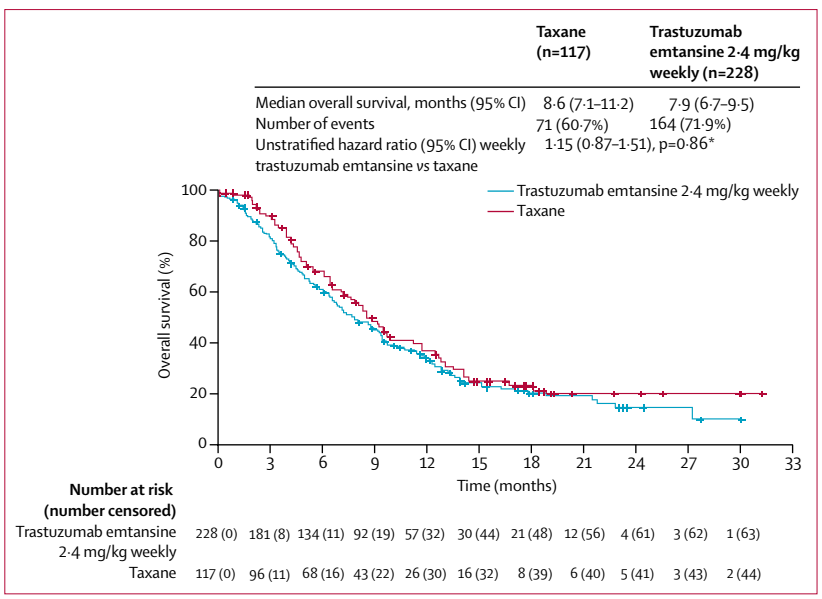

Figure 3: **Kaplan-Meier plot for overall survival in the intention-to-treat population**
*One-sided p value with correction for interim treatment selection in adaptive seamless phase design.

	Overall	Taxane		Trastuzumab emtansine 2·4 mg/kg weekly		Hazard ratio (95% CI)
		Events/ patients	Median (months)	Events/ patients	Median (months)	
Previous HER2-targeted therapy						
Yes	267	54/93	8·9	123/174	8·8	1·12 (0·81–1·55)
No	78	17/24	7·1	41/54	6·8	1·20 (0·68–2·12)
Previous gastrectomy						
Yes	103	19/37	9·0	40/66	11·8	0·94 (0·54–1·62)
No	242	52/80	8·1	124/162	6·6	1·25 (0·90–1·73)
Region						
Asian	157	30/53	11·2	69/104	9·5	1·36 (0·88–2·08)
Non-Asian	188	41/64	6·5	95/124	7·1	0·95 (0·66–1·38)
Age group (years)						
<65	208	38/68	9·7	101/140	7·7	1·34 (0·92–1·95)
≥65	137	33/49	8·0	63/88	8·9	0·92 (0·60–1·40)
Sex						
Female	73	14/22	8·1	36/51	8·7	0·94 (0·50–1·77)
Male	272	57/95	8·6	128/177	7·7	1·21 (0·89–1·66)
Baseline ECOG PS						
0	142	25/43	8·5	69/99	10·7	1·01 (0·64–1·60)
1	201	45/73	8·9	94/128	6·0	1·29 (0·90–1·84)
Extent of disease						
Locally advanced	14	2/4	8·9	8/10	7·1	1·56 (0·31–7·75)
Metastatic	331	69/113	8·5	156/218	8·0	1·14 (0·86–1·51)
Number of lesions at baseline						
1–3	155	30/57	11·2	61/98	10·3	1·14 (0·73–1·76)
>3	190	41/60	6·4	103/130	5·8	1·06 (0·74–1·52)
Visceral (site lung or liver) lesions at baseline						
Yes	219	47/77	7·3	106/142	7·1	1·14 (0·81–1·60)
No	126	24/40	10·0	58/86	8·8	1·20 (0·75–1·93)
Primary site of disease						
Stomach	235	53/84	8·6	106/151	7·9	1·14 (0·82–1·58)
Gastro-oesophageal junction	110	18/33	8·5	58/77	7·1	1·18 (0·70–2·01)
Histological subtype						
Intestinal	137	24/49	12·7	58/88	11·5	1·24 (0·77–2·00)
Diffuse	55	9/12	5·8	32/43	7·7	0·91 (0·43–1·92)
Previous trastuzumab therapy						
Yes	266	54/93	8·9	122/173	8·8	1·12 (0·82–1·55)
No	79	17/24	7·1	42/55	7·2	1·19 (0·68–2·10)
HER2 immunohistochemistry 3+						
Yes	231	49/82	8·5	101/149	9·5	0·99 (0·70–1·40)
No	114	22/35	9·2	63/79	5·2	1·53 (0·94–2·50)
All patients	**345**	**71/117**	**8·6**	**164/228**	**7·9**	**1·15 (0·87–1·51)**

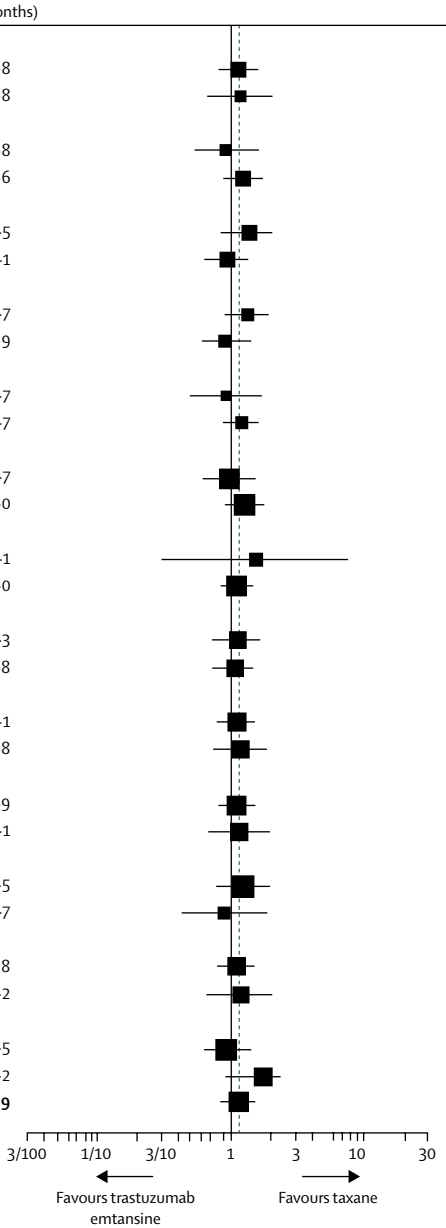

Figure 4: **Forest plot of hazard ratios for overall survival according to patient characteristics at baseline**
ECOG PS=Eastern Cooperative Oncology Group performance status.

Following progression on study treatment, a similar proportion of patients in the trastuzumab emtansine 2·4 mg/kg weekly group as in the taxane group received follow-up anti-cancer treatments (122 [54%] of 228 patients in the trastuzumab emtansine group *vs* 65 [56%] of 117 patients in the taxane group; appendix p 6). A sensitivity analysis for overall survival, which censored patients at the time of initiation of a new treatment, was consistent with the primary analysis and there was no meaningful effect of follow-up cancer treatment (median overall survival was 8·1 months [95% CI 6·8–12·1] in the trastuzumab emtansine 2·4 mg/kg weekly group and 9·4 months [95% CI 7·7–NE] in the taxane group; HR 1·34, 95% CI 0·88–2·05; data not shown).

Among the patients who received at least one study drug, 69 (62%) of 111 patients in the taxane group and 162 (72%)

of 224 patients in the trastuzumab emtansine 2·4 mg/kg weekly group died; 212 deaths were due to progressive disease (60 [54%] of 111 patients in the taxane group *vs* 152 [68%] of 224 patients in the trastuzumab emtansine 2·4 mg/kg weekly group), 12 were due to adverse events (four [4%] of 111 patients in the taxane group *vs* eight [4%] of 224 patients in the trastuzumab emtansine 2·4 mg/kg weekly group), two had unknown causes (two [2%] of 111 patients in the taxane group), and five were classified as died due to other causes (three [3%] of 111 patients in the taxane group *vs* two [1%] of 224 patients in the trastuzumab emtansine 2·4 mg/kg weekly group).

The safety profile of trastuzumab emtansine was consistent with its use in patients with metastatic breast cancer, and no new safety signals were identified (table 3, table 4, appendix pp 2, 7–13).[21–23] Patients in the trastuzumab emtansine 2·4 mg/kg weekly group had a lower incidence of grade 3 or more adverse events (134 [60%] of 224 patients) than those treated with a taxane (78 [70%] of 111 patients; table 3). Grade 3 or more adverse events commonly occurring in patients receiving trastuzumab emtansine were anaemia (59 [26%] of 224 patients) and thrombocytopenia (25 [11%] of 224 patients; table 4, appendix p 2). The most common grade 3 or more adverse events in patients treated with taxanes were neutropenia (43 [39%] of 111 patients) and anaemia (20 [18%] of 111 patients; table 4, appendix p 2). The overall incidence of serious adverse events was similar across the treatment groups (appendix p 11). Commonly occurring serious adverse events in patients receiving 2·4 mg/kg trastuzumab emtansine weekly group were anaemia (8 [4%] of 224 patients) and upper gastrointestinal haemorrhage (8 [4%] of 224 patients). The most common serious adverse events in patients treated in the taxane group were pneumonia (4 [4%] of 111 patients) and febrile neutropenia (4 [4%] of 111 patients).

The proportion of patients discontinuing treatment as a result of an adverse event was similar between groups (31 [14%] of 224 patients in the trastuzumab emtansine 2·4 mg/kg weekly group and 15 [14%] of 111 patients in the taxane group; table 3, appendix p 12). Adverse events leading to discontinuation of at least 1% of patients were pneumonitis (five [2%]), interstitial lung disease (three [1%]), and pneumonia (three [1%]) in the trastuzumab emtansine 2·4 mg/kg weekly group and pneumonitis (three [3%]), peripheral sensory neuropathy (three [3%]), anaphylactic reaction (two [2%]), and hypersensitivity (two [2%]) in the taxane group. Adverse events leading to death were reported in eight (4%) of 224 patients in the trastuzumab emtansine 2·4 mg/kg weekly group (these events were gastric haemorrhage, upper gastrointestinal haemorrhage, atypical pneumonia, septic shock, pneumonia aspiration, and pulmonary alveolar haemorrhage all in one patient each, and pneumonia in two patients) and four (4%) of 111 patients in the taxane group (these events were unexplained cause of death and respiratory failure in one patient each and pneumonia in two patients). Of the four patients with fatal adverse events in the taxane group, one event of pneumonia was assessed by the investigator as being related to the study treatment. Of the eight deaths related to adverse events in the trastuzumab emtansine 2·4 mg/kg weekly group, one

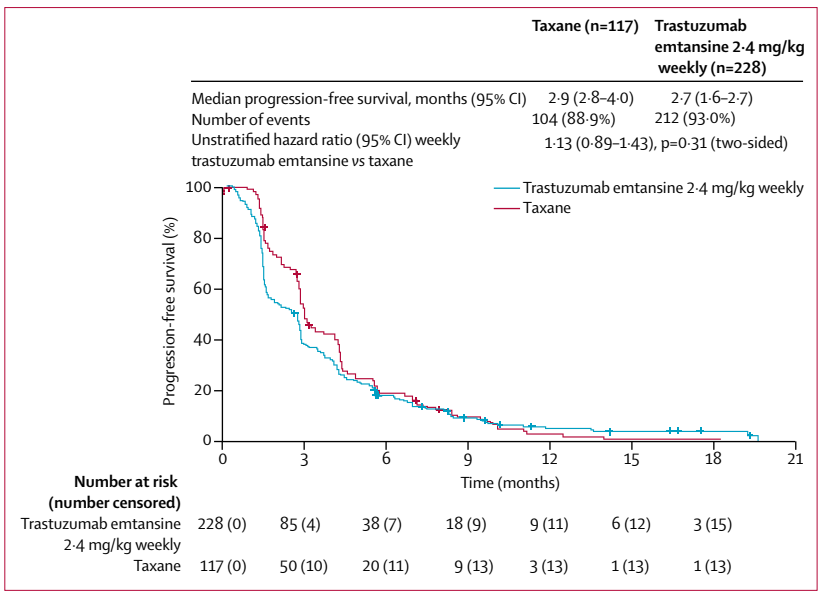

Figure 5: **Kaplan-Meier plot for progression-free survival in the intention-to-treat population**

	Taxane (n=102)	Trastuzumab emtansine 2·4 mg/kg weekly (n=204)	Difference	p value (χ^2)
Objective response	20 (19·6%)	42 (20·6%)	0·98% (−9·04 to 11·00)	0·8406 (two-sided)
Median duration of response* (months)	3·7 (2·8–5·6)	4·3 (3·0–6·8)	..	..

Data are n (%) or (95% CI). Patients with measurable disease at baseline were included in this analysis. *Only patients with an objective response in the measurable disease population were included.

Table 2: **Objective response rate and duration of response in the intention-to-treat population**

	Taxane (n=111)	Trastuzumab emtansine 2·4 mg/kg weekly (n=224)
Any-grade adverse events	108 (97%)	218 (97%)
Grade ≥3 adverse events	78 (70%)	134 (60%)
Serious adverse events	31 (28%)	65 (29%)
Adverse events leading to treatment discontinuation	15 (14%)	31 (14%)
Adverse events leading to death*	4 (4%)	8 (4%)

Data are n (%). *Medical Dictionary for Regulatory Activities-preferred terms include death (unexplained cause), pneumonia (2), and respiratory failure for taxane; and gastric haemorrhage, upper gastrointestinal haemorrhage, pneumonia (2), atypical pneumonia, septic shock, pneumonia aspiration, and pulmonary alveolar haemorrhage for trastuzumab emtansine 2·4 mg/kg weekly.

Table 3: **Overview of adverse events in the safety population**

	Taxane treatment (n=111)	Trastuzumab emtansine 2·4 mg/kg weekly (n=224)
Number of patients with ≥1 grade 3 or higher adverse event	78 (70%)	134 (60%)
Thrombocytopenia*	0	25 (11%)
Anaemia*	20 (18%)	59 (26%)
Hypokalaemia*	0	7 (3%)
Neuropathy peripheral*	0	7 (3%)
Gastric haemorrhage*	2 (2%)	7 (3%)
Upper gastrointestinal haemorrhage*	0	6 (3%)
Asthenia*	0	6 (3%)
Haemorrhage*	4 (4%)	22 (10%)
Gastrointestinal haemorrhage*	1 (1%)	5 (2%)
Aspartate aminotransferase increased*	1 (1%)	5 (2%)
Vomiting*	2 (2%)	5 (2%)
Respiratory tract infection†	3 (3%)	1 (<1%)
Hypoalbuminaemia†	3 (3%)	3 (1%)
Pneumonia†	4 (4%)	7 (3%)
Abdominal pain†	4 (4%)	2 (1%)
Leukopenia†	5 (5%)	0
Decreased appetite†	5 (5%)	4 (2%)
Peripheral sensory neuropathy†	6 (5%)	4 (2%)
Fatigue†	7 (6%)	11 (5%)
Febrile neutropenia†	11 (10%)	2 (1%)
Neutropenia†	43 (39%)	10 (5%)

Data are n (%). Grade 3 or more adverse events by preferred term, with the exception of haemorrhage (grouped adverse event terms per Medical Dictionary for Regulatory Activities Standardised MedDRA Queries); multiple occurrences of the same adverse event in an individual are counted only once. Ordered by incidence rate difference. *Events that favour taxane, †Events that favour trastuzumab emtansine.

Table 4: **Grade 3 or more adverse events with an incidence of 2% or more in either group in the safety population**

event of pulmonary alveolar haemorrhage was assessed by the investigator as being related to study treatment.

Analysis of selected adverse events of interest are shown in the appendix (pp 2, 13). All seven grade 4 or 5 haemorrhagic adverse events were considered by the investigator to have an association with underlying disease; all but the fatal pulmonary alveolar haemorrhage (in one patient) were assessed as being unrelated to study treatment.

Discussion

The randomised phase 2/3 GATSBY study did not show overall survival superiority in patients treated with trastuzumab emtansine 2·4 mg/kg weekly compared with taxane therapy in patients with previously treated HER2-positive advanced gastric cancer. In addition, no clinical or biomarker subgroups were identified that showed an overall survival benefit with trastuzumab emtansine. The results of the secondary efficacy endpoints (progression-free survival, objective response rate, duration of response, and patient-reported outcomes) were consistent with the primary endpoint, even though the secondary endpoints might have been confounded by the effect of the open-label study design on investigator assessment of tumour response and disease progression.

GATSBY had a seamless, adaptive design, allowing for an instantaneous transition to the selected trastuzumab emtansine dose for comparison with taxane therapy. The choice of single-agent taxane (either docetaxel or paclitaxel) as the control group was based on existing treatment guidelines for gastric cancer[25] and a global feasibility study described in the GATSBY study protocol (data not shown). The adaptive, seamless nature of this trial offered potential time and cost savings because, rather than conducting two separate trials, the phase 2 and 3 parts of the trial were separated by interim analyses. Additionally, this seamless design allowed data from patients enrolled into both phases to be included in the final analysis, with an inverse normal combination test used to control fully for type I error. Although this type of study design provided potential for acceleration of the clinical trial process, it is worth noting that additional discussions with health authorities and a greater number of statistical simulations were required because of increased trial complexity. Thus the trial design we used might not be applicable to other settings because early dose selection by an IDMC, and continued treatment in the non-selected treatment group (patients who received 3·6 mg/kg trastuzumab emtansine) after dose selection, might not always be feasible. In this trial, the primary endpoint was overall survival and there were no alternative promising treatment options for patients in this setting.

Despite approvals in second-line gastric cancer therapy,[25] there is an unmet medical need for second-line therapy in patients with HER2-positive advanced gastric cancer who have progressed on or after first-line therapy. Trastuzumab emtansine has previously shown activity in patients with HER2-positive metastatic breast cancer who had progressed during or after HER2-targeted treatment,[21–23] and pre-clinical data suggested that it might also be an effective treatment for HER2-positive gastric cancer (appendix pp 1, 14–16).[20] In GATSBY, demographic and baseline disease characteristics were well balanced, and the study population was deemed representative of the overall population of patients with HER2-positive advanced gastric cancer who had relapsed or progressed during or after a first-line regimen. Overall, a high proportion (44% of patients in the taxane group and 38% of patients in the trastuzumab emtansine group) of patients were missing histological subtype data. GATSBY did not have a mandatory central pathology laboratory to reconcile data discrepancies from different sites and it was also difficult to accurately report Lauren classification from gastric biopsies due to insufficient tumour tissue.

The median overall survival reported with trastuzumab emtansine 2·4 mg/kg weekly (7·9 months, 95% CI 6·7–9·5) was in line with the original study assumption of 9 months; however, the taxane group outperformed the

original assumption set at the time of study design, with a median overall survival that was 2·6 months longer than projected (8·6 months, 7·1–11·2). The original overall survival assumption for the taxane group was based on experimental data from phase 2 trials[14–17,27] of all-comers with second-line advanced gastric cancer; however, data from phase 3 trials[12,13] suggest that this was underestimated, rather than suggesting that patients with more favourable disease biology were recruited to the trial. RAINBOW, a phase 3 study of all-comers with second-line advanced gastric cancer where patients were treated with ramucirumab plus paclitaxel (*vs* placebo plus paclitaxel), reported a median overall survival of 7·4 months (95% CI 6·3–8·4) in the placebo group,[12] whereas TyTAN, a phase 3 study of lapatinib plus paclitaxel (*vs* paclitaxel alone) in Asian patients with HER2-positive advanced gastric cancer, reported a median overall survival of 8·9 months (7·4–11·1) in the paclitaxel group.[13]

Trastuzumab emtansine was well tolerated and discontinuations due to an adverse event as well as the proportion of serious adverse events were similar for patients in both groups (appendix p2). Notably, there was a higher incidence of grade 3 or more pulmonary toxicity, thrombocytopenia, and haemorrhage with trastuzumab emtansine than with taxane treatment, although this was consistent with the known safety profile. Moreover, bleeding events, particularly within the gastrointestinal tract, are also associated with advanced gastric cancer and all seven grade 4 and 5 haemorrhage adverse events were deemed by the investigator to have some association with the underlying disease.

Despite the available preclinical evidence showing trastuzumab emtansine activity in gastric cancer models, emtansine might be less active in gastric cancer; vinca alkaloids such as vinorelbine, which inhibits microtubule polymerisation, seem to be less active in gastric cancer than in other cancers.[28] Additionally, primary and acquired resistance to trastuzumab emtansine have also been observed in preclinical models of gastric cancer.[29] Mechanisms of resistance in preclinical models include multidrug resistance transporters that efflux emtansine out of the cell and decrease HER2 expression.[29,30] Additional resistance mechanisms are postulated to include inefficient internalisation of the HER2–trastuzumab emtansine complex after trastuzumab emtansine treatment and disruption of binding to tubulin due to a β1-tubulin mutation.[29]

In GATSBY, most patients were selected on the basis of archival gastric HER2 status at any disease stage and had previously received a HER2-directed therapy such as trastuzumab. However, after first-line treatment with or without HER2-targeted therapy, there is evidence that HER2 status might be altered or lost during progression because of changes in the molecular profile of the tumour; discordance in HER2 expression has been reported between primary and metastatic tumours[31] and after first-line chemotherapy for metastatic disease with or without trastuzumab.[32,33] Also, HER2 overexpression has a higher incidence of heterogeneity in gastric cancer than in breast cancer.[3] Heterogeneous expression of HER2 might affect activity of trastuzumab emtansine because the nature of the non-cleavable linker does not allow for bystander activity (ie, generation of catabolites that exert cytotoxic activity on neighbouring tumour cells).[34,35] Additionally, focal HER2 overexpression might affect the activity of an antibody–drug conjugate such as trastuzumab emtansine.

Unlike GATSBY, which examined monotherapy in advanced gastric cancer, several studies have investigated the clinical benefit of targeting the HER2 pathway in second line for patients with HER2-positive advanced gastric cancer by using a combination regimen. In the phase 3 study of the dual HER1 and HER2 tyrosine-kinase inhibitor lapatinib combined with paclitaxel (TyTAN), lapatinib plus paclitaxel showed activity in patients with HER2 FISH-positive, immunohistochemistry 3+ gastric cancer, but no significant differences in overall survival in the intention-to-treat population.[13] This trial, however, included Asian patients only, had a low number of patients with previous trastuzumab treatment, and had 35% of patients with a HER2-negative status (immunohistochemistry 0 or 1+).[13]

The HER2 pathway remains a validated target in gastric cancer, and there are several ongoing or planned studies that aim to examine the clinical benefit of disrupting this pathway. The phase 1 part of the TRAX-HER2 study investigates trastuzumab emtansine plus capecitabine in patients with HER2-positive locally advanced or metastatic gastric cancer (NCT01702558). A phase 2 study assessing paclitaxel with the HER1 and HER2 inhibitor afatinib in patients with HER2-positive, trastuzumab refractory advanced oesophagogastric cancer is ongoing (NCT01522768). Several phase 1/2 trials will also examine targeted therapies in combination with immunotherapeutics in this patient population. A phase 1b/2 dose escalation study of margetuximab (fc-optimised chimeric anti-HER2 monoclonal antibody) in combination with pembrolizumab (anti-PD1 antibody) in patients with relapsed or refractory advanced HER2-positive gastro-oesophageal junction or gastric cancer is underway (NCT02689284). Pluripotent killer T cells that are HER2-positive are being investigated for the treatment of advanced gastric cancer in patients with liver metastasis (NCT02632201), and a phase 1/2 clinical trial examines the safety and efficacy of activated autologous natural killer cell infusions after trastuzumab in patients with HER2-positive breast cancer and gastric cancer (NCT02030561).

Overall, the absence of a superiority in overall survival in patients with advanced gastric cancer treated with trastuzumab emtansine compared with those treated with taxanes cannot be explained by imbalances in demographic or baseline disease characteristics, toxicity, or drug administration. Differences in the biology of

HER2-positive gastric cancer and breast cancer, including greater heterogeneity of HER2 expression in gastric cancer, might have contributed to the findings. The results of future clinical trials will clarify whether targeting the HER2 pathway with combination therapy, instead of monotherapy, or with activation of immune cells will improve efficacy after failure of first-line HER2-targeted therapy.

Contributors

PCT-P, MAS, AO, EVC, JAA, HC, KS, GDLP, M-LH-Y, and Y-KK conceived and designed the study. PCT-P, MAS, AO, EVC, JAA, HC, WM, HCC, KS, GDLP, M-LH-Y, BLA, and Y-KK collected data. PCT-P, MAS, AO, HC, HCC, KS, GDLP, TvdH, M-LH-Y, and BLA analysed data. PCT-P, MAS, AO, HC, WM, HCC, GB, KS, GDLP, TvdH, M-LH-Y, BLA, and Y-KK interpreted data. PCT-P, MAS, AO, EVC, JAA, HC, WM, HCC, GB, KS, GDLP, TvdH, M-LH-Y, BLA, and Y-KK wrote, reviewed, or revised the manuscript. All authors approved the final version of the manuscript.

Declaration of interests

PCT-P reports non-financial support from Roche, and personal fees from Roche, Lilly, and Merck Sharp & Dohme, outside of the submitted work. EVC reports research grants from Amgen, Bayer, Boehringer, Celgene, Ipsen, Lilly, Novartis, Merck, Roche, and Sanofi, outside of the submitted work. JAA reports personal fees from Genentech and Roche, outside of the submitted work. HCC reports grant support from Lilly and GSK, and personal fees from Merck-Serono, Lilly, Taiho, Celltrion, Merck Sharp & Dohme, and Quintiles, outside of the submitted work. GDLP, TvdH, MLH-Y, and BLA are employees of F Hoffmann-La Roche and Genentech. TvdH reports shares in F Hoffmann-La Roche. Y-KK reports grant funding from Roche, outside of submitted work. All other authors declare no competing interests.

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